

Casey D. Foley, PhD

Laser & Vacuum Processes | Optical Metrology & Machine Vision | Spectroscopy | Process Development

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SUMMARY

PhD physical chemist and Staff Scientist with 12+ years across production manufacturing, a national laboratory, and university research. Develops and scales laser and high-vacuum processes, with results including 99% yield, 3× faster laser cycles, and a piloted ~40% evacuation cycle-time cut. Builds production metrology and inspection from requirements to plant floor and manages a team of three manufacturing engineers. 23 peer-reviewed publications, including *Science* and *JACS*.

CORE SKILLS

Lasers & optics	Fiber, Nd:YAG/DPSS, CO ₂ , excimer, Ti:sapphire, dye, IR OPO/OPA, VUV; drilling, ablation, scribing, laser heating; optical alignment, beam diagnostics, Zemax OpticStudio, Class 4 laser safety
Vacuum & instruments	Production high vacuum, UHV instrument design, cryogenics, ion optics (SIMION), pulsed discharges, molecular beams, particle imaging, SolidWorks/FreeCAD
Metrology & vision	Phase-shifting deflectometry, confocal profilometry, line-scan machine vision, laser-based R-value measurement, thermal and distortion scanning, GR&R, color measurement (CIELAB), ML surface inspection
Spectroscopy	UV/Vis/IR, FTIR, REMPI, LIF, IR-UV double resonance, velocity-map imaging, mass spectrometry (Waters, Thermo, Bruker), ion mobility
Process & quality	Process development and scale-up, FAT/SAT and commissioning, DOE, SPC, Cp/Cpk and Pp/Ppk, PFMEA, 8D root-cause analysis, control plans, SOPs, preventive maintenance, KPIs, Lean and Six Sigma training, team leadership
Software & data	Python, JMP, LabVIEW, MATLAB; data acquisition, instrument control, machine learning, statistical analysis, automation; data integration with controls and MES teams

PROFESSIONAL EXPERIENCE

Staff Scientist	Apr 2026 – Present
Laser Systems Engineer	Aug 2024 – Mar 2026
<i>LuxWall, Inc. — vacuum-insulated glass manufacturing</i>	

- **Integrated a fiber-laser drilling system:** ran factory acceptance testing and commissioning, cut laser cycle time ~3× at commissioning, eliminated cracking defects, and reduced other defects from ~1 in 10 parts to ≤1% (typical weekly rate) through root-cause analysis, in-line confocal distance sensing, and auto-focus correction.
- **Doubled throughput of a second laser-ablation process at ~99% yield** by integrating in-line confocal depth metrology, closed-loop focus correction, machine-vision part-location correction, and SPC (Ppk, control charts).
- **Own the vacuum-insulated-glass evacuation process,** the factory bottleneck: led its development and commissioning for the first factory, then conceived and led a major process change grounded in desorption kinetics, took it through gated validation (data pipeline, capacity simulator, risk-ordered DOE, FMEA, controlled rollouts), and coordinated it across all levels of the company; piloted changes cut cycle time ~40% while maintaining unit quality.
- **Developed a proprietary laser-based R-value measurement system** from the measurement algorithm through plant-floor integration, and the protocol for using it in production quality screening.
- **Led flatness and distortion metrology:** developed an in-line phase-shifting deflectometry system (~10 μm, ~1M points per panel, automated traceable reports), rebuilt Python control and micron-level analysis for an offline confocal profiler, and wrote analysis software for distortion tools and a thermal scanner.
- **Led specification, vendor selection, and FAT/SAT of an in-line line-scan inspection system** (~50 μm/pixel) that measures panel dimensions and coating edges and detects defects in production; set golden-panel and GR&R acceptance criteria and defined its data requirements with the controls and MES teams.

- **Built a multi-line capacity and bottleneck simulator** that engineering and management use to find each product's constraint and test improvements; rebuilt it from an ~800 MB Python app into sub-1 MB zero-install browser apps with automated parity tests.
- **Manage a team of three manufacturing engineers** (since Apr 2026). Subject-matter expert for laser and high-vacuum processes; defined Class 4 laser-safety controls per ANSI Z136; own control plans, preventive maintenance, SOPs, and work instructions for two production laser systems.

Research Excellence Postdoctoral Fellow

Aug 2022 – Jul 2024

University of Missouri — Department of Chemistry

- **Designed and built a fast-atom/ion beam machine around a pulsed plasma-discharge source** from new and repurposed hardware, toward electron-ion coincidence imaging of associative-ionization reactions relevant to hypersonic flight: Einzel and deceleration optics, quadrupole energy filter, and charge-exchange cell, modeled in SIMION (custom Lua control) and designed in SolidWorks.
- **Developed and characterized pulsed electric-discharge plasma sources over two years:** discharges in a pulsed-valve expansion producing N^+ , N_2^+ , and Ar^+ beams, diagnosed with MCP/phosphor imaging, cameras, and PMTs.
- **Implemented a laser system for VUV detection** of atoms and molecules, generating 118 nm light by frequency-tripling 355 nm in a Xe/Ar gas cell. Performed MeV ultrafast electron diffraction experiments at SLAC National Accelerator Laboratory, with custom Python analysis.
- **Senior postdoc:** mentored and trained group members, and led lab operations and instrument training.

Postdoctoral Associate

Mar 2021 – Jul 2022

Sandia National Laboratories — Combustion Research Facility

- **Returned a relocated \$1M+ cryogenic ion-trap tandem mass spectrometer to full operation in ~7 months:** restored its high-vacuum and 5 K cold-trap systems, aligned and calibrated a tunable IR OPO/OPA (verified against methane photoacoustic spectra) and UV dye lasers, and rebuilt 10 Hz-synchronized timing and scan control with arbitrary waveform generators and LabVIEW.

Graduate Research Assistant (PhD)

May 2016 – Dec 2020

University of Missouri — Suits Group

- **First-author Science paper (2021)** on quantum resonances in formaldehyde photodissociation, measured by velocity-map imaging.

Graduate Research Assistant

Jun 2014 – Apr 2016

Wayne State University

- **Developed ionization and ion-mobility mass spectrometry methods** on Waters, Thermo, and Bruker platforms for biological and polymer analysis; wrote the lab's Synapt calibration SOP and served as laser-safety lead for its Class 3b UV laser.

EDUCATION

PhD, Physical Chemistry, University of Missouri — *Quantum Aspects of Roaming Dynamics* (Prof. Arthur G. Suits) Dec 2020

BS, Biochemistry, Saginaw Valley State University — President's Scholarship; toxicology co-op at The Dow Chemical Company (2010–2012) May 2013

Honors: Army Research Laboratory Journeyman Fellowship (2018) · WSU Thomas C. Rumble Fellowship (2014–2015) · WSU Chemistry Citation for Excellence in Teaching (2014) · SVSU President's Scholarship (2009–2013)

SELECTED PUBLICATIONS

23 peer-reviewed papers (8 first-author) and a 2026 preprint · 450+ citations · h-index 10 · 15+ conference presentations · full list at cdfoley.com/publications

- **C. D. Foley**, C. Xie, H. Guo, A. G. Suits, "Orbiting resonances in formaldehyde reveal coupling of roaming, radical, and molecular channels," *Science* 374, 1122–1127 (2021).
- **C. D. Foley** et al., "Site-specific photochemistry along a protonated peptide scaffold," *J. Am. Chem. Soc.* 146, 13282–13295 (2024).